

Rooftop Solar Potential Mapping of the IIT Delhi Campus

B.Tech Project: Functional Specification and Work Plan, 18 August to 31 October 2026

Supervisor	Prof. Yashasvi [surname], Department of Electrical Engineering	Version	1.0
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1. Purpose and Scope

This document defines the functional behaviour of the analysis system to be built for the project, the acceptance criterion attached to each module, the schedule for the review period from 18 August to 31 October 2026, and the data and approvals required from the supervisor and the Institute. It is submitted as the basis for the first supervisory review and is intended to be marked up and returned.

The project addresses a gap in campus energy planning. A number of flat-roofed buildings at IIT Delhi do not currently host photovoltaic installations, and their generation potential has not been assessed in a systematic, reproducible manner. The system will identify those buildings from high-resolution satellite imagery and open building-footprint data, quantify the usable rooftop area available after obstructions and inter-row shading are accounted for, estimate annual energy yield using measured irradiance data for Delhi, and evaluate the effect of that generation on a representative campus distribution feeder.

Eight to ten candidate buildings will be studied in full, spanning academic blocks, laboratory blocks, hostels and a low-rise service building. Two buildings that already carry rooftop photovoltaic systems will be retained in the sample solely for model validation. Structural load assessment of roofs, procurement and tendering, battery storage sizing, and any on-site instrumentation campaign are outside the scope of this phase.

2. Functional Architecture

The system is organised as five sequential modules with a single shared spatial database. Each module consumes the validated output of the preceding module, writes its results back to the database, and is signed off against the acceptance criterion stated below before the next module begins. This structure is deliberate: it allows a later module to be descoped without invalidating the earlier work, and it makes each intermediate number traceable in the final report.

Ref	Module	Principal function	Output artefact	Acceptance criterion
FR-1	Data acquisition and consolidation	Assemble building footprints from OpenStreetMap and Google Open Buildings, sub-metre imagery, building heights, and hourly irradiance series from PVGIS TMY and NASA POWER for 28.545 N, 77.192 E.	Single project GeoPackage; irradiance comparison note	Every candidate building carries a corrected footprint and a height estimate with a recorded source; the two irradiance sources agree on annual global horizontal irradiation within 5 per cent.
FR-2	Roof characterisation and usable area	Manual digitisation of roof outlines, parapets and obstructions in QGIS, supported by a segmentation network trained on the INRIA aerial dataset and a classifier that flags roofs already carrying panels.	Usable-area register; segmentation accuracy report	Segmentation intersection over union of at least 0.75 against the manually digitised ground truth; usable area derived from a documented setback and ground coverage ratio rather than an assumed percentage.
FR-3	Shading analysis	Extrude footprints to a digital surface model, compute annual sun position, and derive an hourly shading factor for each roof from adjacent buildings, parapets and tree canopy.	Hourly shading factor time series per roof	Modelled shadow geometry reproduces a shadow observed in dated satellite imagery to within 15 per cent of its measured length.
FR-4	Generation estimation	Per-building pvlib model chain using Perez transposition, an explicit cell-temperature model and an itemised loss stack covering soiling, mismatch, wiring, clipping and availability.	Annual energy, specific yield and capacity factor per building	Specific yield falls within 1300 to 1600 kWh per kWp per year and lies within 10 per cent of the PVGIS estimate for a reference building; back-tested against metered output of an existing campus installation.
FR-5	Feeder impact study	Representative 11 kV feeder modelled in OpenDSS with annual time-series power flow, evaluating steady-state voltage rise, reverse power flow at distribution transformers, feeder losses and hosting capacity.	Feeder model, annual results, hosting capacity limit	Base-case transformer loading reproduces billed consumption within 15 per cent; all limits assessed against the applicable CEA regulations and IEEE 1547 practice.

3. Schedule and Review Points

Phase	Period	Tasks	Milestone and review
A Setup	Wk 1 to 2 18 to 31 Aug	Scope lock with supervisor; repository and environment set up; data requests issued to the Estate Office and the Electrical Maintenance Division; literature review of rooftop potential estimation methods completed.	M1 , 31 Aug. Building list frozen; data requests acknowledged. Review meeting.

Phase	Period	Tasks	Milestone and review
B Data and roofs	Wk 3 to 5 1 to 21 Sep	FR-1 delivered. Manual digitisation of all candidate roofs and obstructions; segmentation model trained and evaluated; usable-area register produced with the inter-row spacing derivation.	M2 , 21 Sep. Usable area per building agreed. Mid-term review; interim presentation if required.
C Yield	Wk 6 to 8 22 Sep to 12 Oct	FR-3 and FR-4 delivered. Shading factors computed, generation model run for all candidate buildings, validation against existing installations and PVGIS completed, campus heat map produced.	M3 , 12 Oct. Validated annual yield figures. Review meeting.
D Grid	Wk 9 to 10 13 to 26 Oct	FR-5 delivered. Feeder assembled and calibrated, load profiles constructed, annual time-series study run, hosting capacity determined for the studied feeder.	M4 , 26 Oct. Grid impact results. Review meeting.
E Synthesis	Wk 11 27 Oct to 2 Nov	Ranked building table, campus-level capacity estimate, indicative techno-economic assessment and avoided emissions, consolidated interim report and code documentation.	M5 , 2 Nov. Interim report submitted for comment.

A standing fortnightly slot of thirty minutes is requested in addition to the four milestone reviews, so that method choices can be corrected early rather than at the milestone itself.

4. Support and Data Requested from the Supervisor

Item	Purpose	Required by
Confirmation of the candidate building list, and whether a campus-wide extrapolation is expected in addition to the detailed sample	Locks the scope of FR-1 and FR-2	Week 1
Departmental letter or introduction to the Estate Office and the Electrical Maintenance Division	Releases the campus single-line diagram, transformer schedule and feeder layout needed for FR-5	Week 2
Monthly electricity consumption records, or meter data where available, for the candidate buildings	Construction of realistic load profiles and calibration of the base case	Week 3
Generation records for the existing campus rooftop installations	Independent validation of FR-4; without this the yield model can only be cross-checked against PVGIS	Week 4
Guidance on the relative weight expected between the machine-learning component and the power system study	Allocation of effort between FR-2 and FR-5	Week 2
Access to a licensed tool such as PVsyst or ETAP, if the Department holds one	Optional cross-verification of yield and load-flow results	Week 6
Direction on whether a conference or journal submission is to be targeted from this work	Determines the level of documentation, benchmarking and novelty framing carried through Phase E	Week 4
Permission for supervised roof access on two or three buildings	Ground-truthing of obstructions that are ambiguous in imagery	Week 4

5. Deliverables at the End of the Review Period

The period closes with a validated spatial database of campus roofs; a campus heat map of annual specific yield; a ranked table of untapped buildings by installable capacity and by yield; a campus-level estimate of untapped capacity in MWp and annual energy in MWh; a hosting-capacity figure for the studied feeder; an indicative payback and levelised cost of energy at the prevailing campus tariff; and a documented, version-controlled codebase that reproduces every figure in the report from the raw data.

6. Risks and Mitigation

Risk	Consequence	Mitigation
Estate Office data is delayed or withheld	FR-5 cannot use the true network	Proceed with a representative feeder built from transformer ratings and distances measured off the campus map, with all assumptions declared; escalate through the supervisor in Week 4 if no response.
Imagery resolution insufficient for small rooftop obstructions	Usable area overstated	Ground-truth two or three roofs on foot and apply the observed correction factor to the remainder, reporting the residual uncertainty as a band rather than a point estimate.
Segmentation model underperforms on limited local training data	FR-2 acceptance criterion missed	Manual digitisation remains the primary source of area; the model is then reported as a comparative transfer-learning study with a stated failure analysis, which preserves the contribution.
Phase C or D overruns	Synthesis compressed	Reduce full shading analysis to four buildings and apply a validated flat-roof shading factor to the rest; Phase E is protected and not compressed.

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Signature and date:

Signature and date: